OLIST BUSINESS ANALYTICS

Data Cleaning

Phase Name and Number	Task	Output	Insights
Phase 1 Database Creation	Create a database and import all required CSV files	Data imported Successfully And Verification Each table with select done	These three Phases help to understand data and how to work with it in future phases in processes of business analysis.
	After importing the data, verify every table		
Phase 2 Data Understanding	Understand the dataset by find the total number of orders, customers, unique customers, products, sellers, product categories, cities, and states…etc..	Understand the data is done and implemented in different phases	
Phase 3 Data Quality Analysis	Missing Values: Identify NULL values in important columns such as delivery dates, approval dates, product categories, review scores, and customer information.	Identified null values and duplicates and without changing any-data, implemented measures in different phases to ignore or prevent data accuracy	
	Duplicate Records: Check duplicate IDs or records for order_id, customer_id, product_id, seller_id, and review_id. Do not automatically delete duplicates.		

OLIST BUSINESS ANALYTICS

Phase-4: Order Analysis

Topic	Task	Output	File name	Dataset used
Order analysis	1. How many total orders were placed?	99441	Order analysis	olist_orders_dataset
	2.How many orders were delivered?	96478	Order analysis	olist_orders_dataset
	3.How many orders were cancelled?	625	Order analysis	olist_orders_dataset
	4. What percentage of orders were successfully delivered?	97.02%	Order analysis	olist_orders_dataset
	5. What percentage of orders were cancelled?	0.63%	Order analysis	olist_orders_dataset
	6.What are the different order statuses?	Delivered, invoiced, shipped, processing, unavailable, canceled, created, approved	Order analysis	olist_orders_dataset
	7. How many orders belong to each status?	Delivered-96478, Invoiced-314, shipped-1107, processing-301, …	Order analysis	olist_orders_dataset

Phase-5: Sales Analysis

Topic	Task	Output	File name	Dataset used
Sales Analysis	1. What is the total product sales value?	13591644	Sales Analysis	olist_order_items_dataset
	2. What is the total freight value?	2251909.54	Sales Analysis	olist_order_items_dataset
	3. What is the total revenue including freight?	15843553.24	Sales Analysis	olist_order_items_dataset
	4. What is the average order value?	121	Sales Analysis	olist_order_items_dataset
	5. What is the highest-value order?	6735	Sales Analysis	olist_order_items_dataset
	6. What is the lowest-value order?	0.85	Sales Analysis	olist_order_items_dataset
	7. How much revenue is generated each year?	year, revenue_of_year 2017, 7000608.86 2018, 8785262.97 2016, 57183.21 2020, 498.2	Sales Analysis	olist_order_items_dataset
	8. How much revenue is generated each month?	month, year, revenue_of_month 10, 2016, 56945.07 9, 2016, 218.52 12, 2016, 19.62 12, 2017, 1045817.04 11, 2017, 1027375.7	Sales Analysis	olist_order_items_dataset
	9. Which month generated the highest revenue?	month, year, revenue_of_month 10, 2016, 56945.07 12, 2017, 1045817.04 8, 2018, 1259175.95 4, 2020, 322.86	Sales Analysis	olist_order_items_dataset
	10. Which month generated the lowest revenue?	month, year, revenue_of_month 12, 2016, 19.62 1, 2017, 92198.41 9, 2018, 16026.36 2, 2020, 175.34	Sales Analysis	olist_order_items_dataset

OLIST BUSINESS ANALYTICS

Phase-6: Sales Growth Analysis

<table>
<tr><th>Topic</th><th>Task</th><th>Output</th><th>File name</th><th>Dataset used</th></tr>
<tr><td rowspan="4">Sales Growth Analysis</td><td>1. Calculate monthly revenue</td><td><table><tr><th>month</th><th>year</th><th>revenue_of_month</th></tr><tr><td>10</td><td>2016</td><td>56945.07</td></tr><tr><td>9</td><td>2016</td><td>218.52</td></tr><tr><td>12</td><td>2016</td><td>19.62</td></tr><tr><td>12</td><td>2017</td><td>1045817.04</td></tr></table></td><td>Sales Growth Analysis</td><td>olist_order_items_dataset</td></tr>
<tr><td>2. Calculate Previous month's</td><td><table><tr><th>month</th><th>year</th><th>month_revenue</th><th>previous_month_revenue</th></tr><tr><td>9</td><td>2016</td><td>218.52</td><td>NULL</td></tr><tr><td>10</td><td>2016</td><td>56945.07</td><td>218.52</td></tr><tr><td>12</td><td>2016</td><td>19.62</td><td>56945.07</td></tr><tr><td>1</td><td>2017</td><td>92198.41</td><td>19.62</td></tr></table></td><td>Sales Growth Analysis</td><td>olist_order_items_dataset</td></tr>
<tr><td>3.Calculate Revenue difference</td><td><table><tr><th>month</th><th>year</th><th>month_revenue</th><th>previous_month_revenue</th><th>differnce_in_revenue</th></tr><tr><td>9</td><td>2016</td><td>218.52</td><td>NULL</td><td>NULL</td></tr><tr><td>10</td><td>2016</td><td>56945.07</td><td>218.52</td><td>56726.55</td></tr><tr><td>12</td><td>2016</td><td>19.62</td><td>56945.07</td><td>-56925.45</td></tr><tr><td>1</td><td>2017</td><td>92198.41</td><td>19.62</td><td>92178.79</td></tr></table></td><td>Sales Growth Analysis</td><td>olist_order_items_dataset</td></tr>
<tr><td>4. Calculate month-over-month growth percentage</td><td><table><tr><th>month1</th><th>ar</th><th>mr</th><th>percentage</th></tr><tr><td>1</td><td>7383995.351671453</td><td>7506643.240004479</td><td>98.37</td></tr><tr><td>2</td><td>7208020.394791371</td><td>7506643.240004479</td><td>96.02</td></tr><tr><td>3</td><td>7154844.466031301</td><td>7506643.240004479</td><td>95.31</td></tr><tr><td>4</td><td>7157284.322707757</td><td>7506643.240004479</td><td>95.35</td></tr></table></td><td>Sales Growth Analysis</td><td>olist_order_items_dataset</td></tr>
</table>

Phase-7: Customer Analysis

<table>
<tr><th>Topic</th><th>Task</th><th>Output</th><th>File name</th><th>Dataset used</th></tr>
<tr><td rowspan="8">Customer Analysis</td><td>1. How many unique customers exist?</td><td>96096</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>2. Which state has the highest number of customers?</td><td>SP - 41746</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>3. Which city has the highest number of customers?</td><td>sao paulo - 15540</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>4. What are the top 10 cities by customer count?</td><td><table><tr><th>customer_city</th><th>customer_count</th></tr><tr><td>sao paulo</td><td>15540</td></tr><tr><td>rio de janeiro</td><td>6882</td></tr><tr><td>belo horizonte</td><td>2773</td></tr><tr><td>brasilia</td><td>2131</td></tr></table></td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>5. How many customers placed more than one order?</td><td>0</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>6. What percentage of customers are repeat customers?</td><td>0</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>7. Which customers placed the most orders?</td><td>All records in table</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
<tr><td>8. Which customers generated the highest revenue?</td><td>Customer_Id-13664.08</td><td>Customer Analysis</td><td>olist_customers_dataset</td></tr>
</table>

Phase-8: Customer Segmentation

Topic	Task	Output	File name	Dataset used
Customer Segmentation	1. Segment customers based on spending into Low Value, Medium Value, High Value, and VIP customers. Students may define appropriate thresholds.	Column added and Updated with records	Customer Segmentation	olist_order_payments_dataset

OLIST BUSINESS ANALYTICS

Phase-9: Product Analysis

Topic	Task	Output	File name	Dataset used
Product Analysis	1. How many unique products were sold?	product_category_name - sold cama_mesa_banho 11115 beleza_saude 9670 esporte_lazer 8641 moveis_decoracao 8334	Product Analysis	olist_products_dataset
	2. How many product categories exist?	73	Product Analysis	olist_products_dataset
	3. Which categories have the highest number of products?	cama_mesa_banho-3029	Product Analysis	olist_products_dataset
	4. Which products sold the most units?	Product_Id- 527	Product Analysis	olist_products_dataset
	5. Which product categories sold the most units?	cama_mesa_banho-11115	Product Analysis	olist_products_dataset
	6. Which categories generated the highest revenue?	beleza_saude -1441248.07	Product Analysis	olist_products_dataset
	7. Which categories generated the lowest revenue?	seguros_e_servicos-324.51	Product Analysis	olist_products_dataset
	8. What is the average product price by category?	product_category_name - average_product_price pcs 1146.8 portateis_casa_forno_e_cafe 660.44 eletrodomesticos_2 520.66 agro_industria_e_comercio 369.69	Product Analysis	olist_products_dataset
	9. What are the top 10 product categories by revenue?	product_category_name - sold - revenue beleza_saude 9670 1441248.07 relogios_presentes 5991 1305541.61 cama_mesa_banho 11115 1241681.72 esporte_lazer 8641 1156656.48	Product Analysis	olist_products_dataset
	10. What percentage of total revenue comes from the top 10 categories?	63	Product Analysis	olist_products_dataset

Phase-10: Product Ranking

Topic	Task	Output	File name	Dataset used
Product Ranking	1. Using Rank products based on sales. Use ROW_NUMBER (), RANK (), and DENSE_RANK (). Find the top 5 products by revenue in each product category.	Each categories top 5 with revenue and number of sold	Product Ranking	olist_products_dataset

OLIST BUSINESS ANALYTICS

Phase-11: Seller Analysis

Topic	Task	Output	File name	Dataset used
Seller Analysis	1. How many sellers are active?	3095	Seller Analysis	olist_sellers_dataset olist_order_items_dataset
	2. Which state contains the highest number of sellers?	SP-1849	Seller Analysis	olist_sellers_dataset
	3. Which city contains the highest number of sellers?	Sao paulo-694	Seller Analysis	olist_sellers_dataset
	4. Which sellers processed the most orders?	Seller_id-2033	Seller Analysis	olist_sellers_dataset olist_order_items_dataset
	5. Which sellers sold the most products?	Seller_id-2033	Seller Analysis	olist_sellers_dataset olist_order_items_dataset
	6.Which sellers generated the highest revenue?	Seller_id-151265.77	Seller Analysis	olist_sellers_dataset olist_order_items_dataset
	7. What is the average order value for each seller?	seller_id / orders_processed / avg_order_value 6560211a19b47992c3666cc44a7e94c0 2033 74.41 4a3ca9315b744ce9f8e9374361493884 1987 118.54 1f50f920176fa81dab994f9023523100 1931 73.59	Seller Analysis	olist_sellers_dataset olist_order_items_dataset
	8. Who are the top 10 sellers by revenue?	seller_id / orders_processed / revenue 4869f7a5dfa277a7dca6462dcf3b52b2 1156 249640.7 7c67e1448b00f6e969d365cea6b010ab 1364 239536.44 53243585a1d6dc2643021fd1853d8905 410 235856.68	Seller Analysis	olist_sellers_dataset olist_order_items_dataset

Phase-12: Payment Analysis

Topic	Task	Output	File name	Dataset used
Payment Analysis	1. What payment methods are available?	credit_card, boleto , voucher, debit_card	Payment Analysis	olist_order_payments_dataset
	2. Which payment method is used most frequently?	credit_card-76795	Payment Analysis	olist_order_payments_dataset
	3. How much revenue comes through each payment method?	credit_card 12542084.19 boleto 2869361.27 voucher 379436.87 debit_card 217989.79	Payment Analysis	olist_order_payments_dataset
	4. What percentage of payments belong to each payment type?	payment_type / revenue / revenue_percentage credit_card 12542084.19 78 boleto 2869361.27 18 voucher 379436.87 2 debit_card 217989.79 1	Payment Analysis	olist_order_payments_dataset
	5. What is the average payment value by payment type?	credit_card 163.3 boleto 145 debit_card 142.6 voucher 65.7	Payment Analysis	olist_order_payments_dataset
	6. What is the maximum number of instalments used?	payment_type / maxium_installments credit_card 24 boleto 1 voucher 1	Payment Analysis	olist_order_payments_dataset
	7. What is the average number of instalments?	payment_type / avg_installments_used credit_card 3.51 boleto 1.00 voucher 1.00	Payment Analysis	olist_order_payments_dataset
	8. How does average transaction value differ between payment methods?	payment_type / avg_transaction_value credit_card 163.3190206393595 boleto 145.03443540234454 debit_card 142.57017004578148	Payment Analysis	olist_order_payments_dataset

OLIST BUSINESS ANALYTICS

Phase-13: Delivery Performance

Topic	Task	Output	File name	Dataset used
Delivery Performance	1. What is the average delivery time?	12.54 days/ 300.90 hrs	Delivery Performance	olist_orders_dataset
	2. What is the minimum delivery time?	12 hrs /0.50 day	Delivery Performance	olist_orders_dataset
	3. What is the maximum delivery time?	5031 hrs / 209.63 days	Delivery Performance	olist_orders_dataset
	4. Which states have the fastest delivery?	SP-209.77 hrs	Delivery Performance	olist_orders_dataset olist_customers_dataset
	5. Which states have the slowest delivery?	RR-704.73 hrs	Delivery Performance	olist_orders_dataset olist_customers_dataset
	6. Which cities experience the longest delivery times?	Novo brasil-3556.00 hrs	Delivery Performance	olist_orders_dataset olist_customers_dataset
	7. How many orders arrived before the estimated delivery date?	88652	Delivery Performance	olist_orders_dataset
	8. How many orders arrived after the estimated delivery date?	7826	Delivery Performance	olist_orders_dataset
	9. What percentage of orders were delivered late?	On Time-92 Late-8	Delivery Performance	olist_orders_dataset

Phase-14: Customer Review Analysis

Topic	Task	Output	File name	Dataset used
Customer Review Analysis	1. What is the average review score?	4.01	Customer Review Analysis	olist_order_reviews_dataset
	2. How many reviews received each rating from 1 to 5?	review_score / count_rating 5 / 164 4 / 39 1 / 38 3 / 30 2 / 8	Customer Review Analysis	olist_order_reviews_dataset
	3. What percentage of reviews are positive?	86	Customer Review Analysis	olist_order_reviews_dataset
	4. What percentage are negative?	14	Customer Review Analysis	olist_order_reviews_dataset
	5. Which product categories have the highest average ratings?	informatica_acessorios- 21 4.43 21	Customer Review Analysis	olist_order_reviews_dataset olist_products_dataset olist_order_items_dataset
	6. Which sellers generated the highest revenue?	utilidades_domesticas- 31 3.55 31	Customer Review Analysis	olist_order_reviews_dataset olist_products_dataset olist_order_items_dataset
	7. Which sellers have the highest average ratings?	Seller_id – 6 5.00 6	Customer Review Analysis	olist_order_reviews_dataset olist_sellers_dataset olist_order_items_dataset
	8. Which sellers have the lowest average ratings?	Seller_id – 6 1.00 6	Customer Review Analysis	olist_order_reviews_dataset olist_sellers_dataset olist_order_items_dataset

OLIST BUSINESS ANALYTICS

Phase-15: Delivery vs Customer Satisfaction

Topic	Task	Output	File name	Dataset used
Delivery vs Customer Satisfaction	1.Compare order counts and average review score for On Time and Late deliveries	arrived / avg_score On Time 4.2398 Late 2.5000	Customer Review Analysis	olist_orders_dataset olist_order_reviews_dataset

Phase-16: Freight Cost Analysis

Topic	Task	Output	File name	Dataset used
Freight Cost Analysis	1. What is the total freight value?	2251909.54	Freight Cost Analysis	olist_order_items_dataset
	2. What is the average freight value per order?	Order_id – 5 0	Freight Cost Analysis	olist_order_items_dataset
	3. Which states have the highest freight costs?	RO-50.91	Freight Cost Analysis	olist_order_items_dataset
	4. Which product categories have the highest average freight costs?	Pcs -48.45	Freight Cost Analysis	olist_order_items_dataset
	5. Which products have unusually high freight charges?	Product – 409.68	Freight Cost Analysis	olist_order_items_dataset

OLIST BUSINESS ANALYTICS

Insights

Phase Number	Phase Name	INSIGHT
4.	Order Analysis	The total orders are around 99K, with 97.02% successfully delivered. Cancelled orders are only 0.63%, showing a very low cancellation rate. Overall, the order process is performing efficiently.
5.	Sales Analysis	Total revenue including freight is around 15.84 million. Product sales contribute the major portion, while freight adds around 2.25 million. This helps understand the overall revenue contribution and order value.
6.	Sales Growth Analysis	Monthly revenue and growth percentages help identify increasing and decreasing sales periods. It shows how revenue changes from one month to another. This supports better sales planning and forecasting.
7.	Customer Analysis	There are around 96K unique customers in the dataset. São Paulo has the highest customer concentration with around 41K customers. This shows strong customer demand in major regions.
8.	Customer Segmentation	Customers are divided by spending levels to identify valuable customers and support targeted retention strategies.
9.	Product Analysis	There are 73 product categories in the dataset. Cama_mesa_banho has the highest product volume, while beleza_saude generates the highest revenue. This shows differences between product availability and revenue performance.
10.	Product Ranking	Ranking identifies the top products within each category, helping with inventory and product-performance decisions.
11.	Seller Analysis	There are around 3,095 active sellers, with the highest concentration in São Paulo. Some sellers contribute significantly to orders, product sales, and revenue. This helps identify high-performing sellers and areas needing support.
12.	Payment Analysis	There are 4 payment methods, with credit card being the most frequently used. This shows the strong preference for card-based payments. The business can use this to improve the checkout experience.
13.	Delivery Performance	The average delivery time is around 12.54 days, while about 8% of orders were late. São Paulo has faster delivery, while some regions experience much longer delivery times. This highlights opportunities to improve regional logistics.
14.	Customer Review Analysis	The average customer rating is 4.01 out of 5. Around 86% of reviews are positive, showing generally good customer satisfaction. Low-rated products and sellers can be targeted for improvement.
15.	Delivery vs Customer Satisfaction	Comparing late and on-time deliveries with review scores helps understand their effect on satisfaction. It can identify whether delivery delays lead to lower customer ratings. This connects logistics performance with customer experience.
16.	Freight Cost Analysis	Total freight value is around 2.25 million, making shipping an important cost factor. Some states and product categories have higher freight costs than others. This highlights opportunities to optimize shipping and reduce logistics costs.